

DIFFERENTIALLY PRIVATE MEDIAN OF MEANS AND TRIMMED MEAN

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ABSTRACT

We explore ways to construct differentially private one-dimensional mean estimators from classic robust ones. We introduce and implement private versions of the median of means and the trimmed mean using a recent private quantile estimation algorithm. We empirically demonstrate PRIVMOM’s superior performance against other private mean estimators on synthetic data. Skeptical of our algorithm’s dramatic improvement in error rate, we also conduct membership inference hypothesis testing to ensure that its performance does not exceed theoretically achievable bounds.

1 INTRODUCTION

Mean estimation is a fundamental statistical task, and it is nontrivial to do accurately while preserving data privacy. In this paper, we study the problem of estimating the mean μ of a distribution from which we obtain N i.i.d. samples while guaranteeing the privacy of points in the dataset. Differential privacy provides a mathematical framework to guarantee that the output of an algorithm—in our case, a mean estimator—does not reveal too much about any single data point. In other words, we would like to limit how sensitive the algorithm is to individual points.

A similar idea underpins the field of robust statistics, whose aim is to derive statistical estimators that are robust to a specified fraction of “contaminated” data (and hence to the addition or removal of one data point). For instance, the sample median is a robust estimator, but the mean is not.

Much work has been done relating statistical robustness and differential privacy, including deriving private estimators from robust ones. In this paper, we continue this line of work with two robust estimators made private. Our contributions are as follows:

1. We present the Private Median of Means (PRIVMOM) and Private Trimmed Mean (PRIVTM) mean estimation algorithms.
2. We compare their performance against other mean estimation algorithms.
3. We attempt to check that PRIVMOM does not exceed theoretical bounds by conducting membership inference hypothesis tests.

1.1 RELATED WORK

The median of means is a well-known robust and highly accurate mean estimator; its error decays exponentially in the number of samples (assuming finite variance). It was first introduced by Nemirovsky & Yudin (1983) and Alon et al. (1999). More recently, Lugosi & Mendelson (2021) prove that another classic robust estimator, the trimmed mean, can also attain similar (so-called *subgaussian*) rates with properly chosen clipping thresholds.

Differential Privacy was developed in a series of celebrated works by Dwork, McSherry, Nissim, Smith, and other collaborators in the 2000s. For the problem of private mean estimation, Cai et al. (2021) show that no ϵ -private mean estimator can attain an error rate better than $O(1/\sqrt{n} + 1/n\epsilon)$.

The Sparse Vector Technique was originally introduced in Dwork et al. (2009), and its instantiation as the ABOVETHRESHOLD algorithm is given in Chapter 3 of the Differential Privacy textbook

Dwork et al. (2014). The textbook also provides a basic private histogram algorithm, which can be used to estimate a bounded range within which the mean falls. It is used by some mean estimation algorithms to calibrate the noise.

More recently, Durfee (2023) presented a new algorithm based on the Sparse Vector Technique to privately estimate quantiles of a given dataset with a more efficient use of ϵ . The Sparse Vector Technique takes advantage of the fact that the privacy budget is only spent on a few, sparse queries. We use this as a subroutine to estimate the median in our median of means algorithm as well as the prescribed clipping thresholds in our trimmed mean algorithm.

Much work has explored connections between privacy and robustness. Dwork & Lei (2009) first showed how robust estimators can produce differentially private ones using their Propose-Test-Release paradigm. Concurrent works by Hopkins et al. (2023) and Asi et al. (2023) even showed that any contamination-robust algorithm could be made private, albeit in a computationally inefficient manner. See Kamath (2025) for an extended discussion on private and robust mean estimation.

Kamath et al. (2020) proposed a private mean estimation algorithm for heavy-tailed data with k bounded moments. We use their one-dimensional version, PDPODME, as a baseline for comparison. Liu et al. (2021) propose PRIME, a robust and differentially private algorithm for high-dimensional mean estimation, which we also include for comparison.

2 METHODS

Our proposed algorithms rely heavily on Durfee (2023)’s unbounded quantile estimation algorithm. It starts at a lower bound ℓ and searches for the qN th largest entry by repeatedly calling the well-known differentially private ABOVETHRESHOLD algorithm with the query

$$f_i(X) = |\{x_j \in X : x_j - \ell + 1 < \beta^i\}|.$$

It queries the number of data points that lie below the slowly increasing candidate quantile β^i until the (noisy) $f_i(X)$ exceeds qN . β is typically initialized at $\beta = 1.001$ but can be adjusted according to the use case¹. PRIVATEQUANTILE and its subroutines can be found in Appendix A

We compare our results to Algorithm 1 (PDPODME) from (Kamath et al., 2020), which constructs a private histogram using the Laplace mechanism, then uses the bin with the (noisy) highest count to set a bound on the range within which the mean lies. It returns a noisy mean with noise calibrated to the bounded range of the bin.

2.1 ALGORITHMS

We now present our main two algorithms. The PRIVATEMEDIANOFMEANS algorithm partitions X into B blocks, calculates block means $\hat{\mu}_b$, and returns the noisy median of the $\hat{\mu}_b$ ’s by calling PRIVATEQUANTILE to retrieve the noisy 50th percentile. PRIVATEMEDIANOFMEANS is automatically private by the privacy of PRIVATEQUANTILE, which, in turn, is private by its call to ABOVETHRESHOLD.

Algorithm 1 PRIVATE MEDIAN OF MEANS (PRIVMOM)

- 1: **Input:** 1-dimensional data vector X , number of blocks B , privacy budget ϵ
 - 2: Randomly partition X into B blocks of length L
 - 3: **for** $b = 1$ to B **do**
 - 4: Compute $\hat{\mu}_b \leftarrow \frac{1}{L} \sum_{l=1}^L X_l^b$
 - 5: **end for**
 - 6: Compute $\hat{\mu} \leftarrow \text{PRIVATEQUANTILE}(\{\hat{\mu}_1, \dots, \hat{\mu}_B\}, q = 0.5, \epsilon/2, \epsilon/2)$
 - 7: **return** $\hat{\mu}$
-

Our second algorithm, PRIVATETRIMMEDMEAN, is a noisy version of the optimal one-dimensional trimmed mean proposed by Lugosi & Mendelson (2021). We compute the noisy trimming thresholds using the PRIVATEQUANTILE subroutine, then return the trimmed mean with Laplace noise scaled to the width of the trimming interval.

¹Here β grows exponentially, but PRIVATEQUANTILE can also be modified to search at uniform intervals.

Algorithm 2 PRIVATE TRIMMED MEAN (PRIVTM)

- 1: **Input:** X containing N data points, privacy budget ϵ , confidence level $0 < \rho < 1$
 - 2: $\epsilon' \leftarrow 12 \frac{\log(4/\rho)}{N}$
 - 3: Randomly partition X into Y, Z each of length $N/2$
 - 4: $l \leftarrow \text{PRIVATEQUANTILE}(Y, q = \epsilon', \epsilon/5, \epsilon/5)$
 - 5: $r \leftarrow \text{PRIVATEQUANTILE}(Y, q = 1 - \epsilon', \epsilon/5, \epsilon/5)$
 - 6: Compute $\hat{\mu} \leftarrow \frac{2}{N} \sum_{i=1}^{N/2} \text{Trim}(Z_i, l, r)$
 - 7: **return** $\hat{\mu} + \text{Lap}\left(\frac{10(r-l)}{N\epsilon}\right)$
-

3 RESULTS

In this section, we compare the performance of our proposed algorithms to those of others in the literature. We then conduct membership inference testing to check that PRIVMOM does not exceed theoretical bounds. We make our code publically available on GitHub.

3.1 EXPERIMENTS

We tune the hyperparameters of PRIVATEMEDIANOFMEANS and PRIVMOM in Appendix B. We compare the performance of PRIVATEMEDIANOFMEANS, PRIVATETRIMMEDMEAN, PRIME, PDPODME as N and ϵ vary².

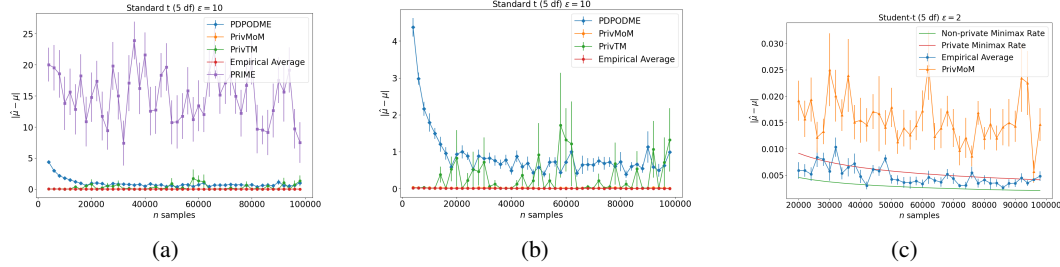


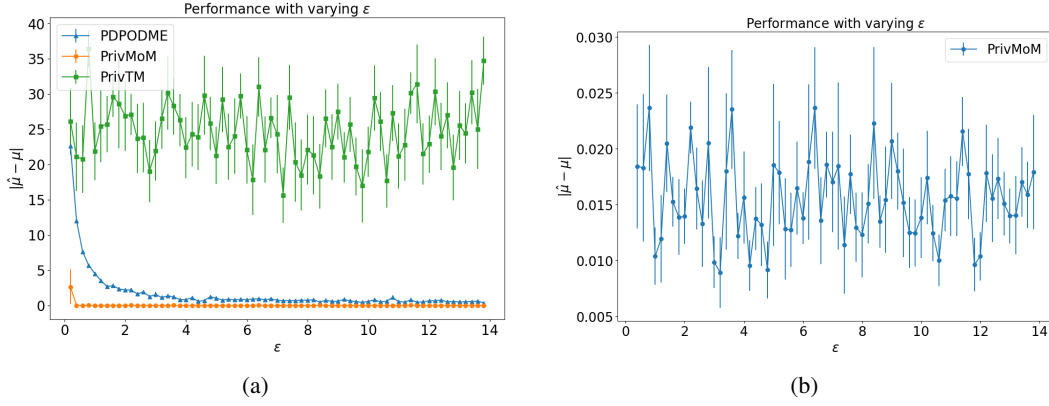
Figure 1: Performance of mean estimators with varying sample size. (a) is the same as (b) but with PRIME included for completeness. (c) plots the minimax rates $O(\sigma/\sqrt{n})$ and $O(\sigma/\sqrt{n} + \sigma/n\epsilon)$, $\sigma = \sqrt{5/3}$ ³. The figures plot average error over 10 trials, each trial sampling N i.i.d. points from Student-t distributions with $\text{df}=5$ and means randomly selected between -50 and 50.

3.2 DISCUSSION

It can be seen in Figure 1b that as N grows, PDPODME and PRIVTM attain similar error rates, but PRIVMOM vastly outperforms both of them. In Figure 1a PRIME understandably performs worse, as it is meant to be a high-dimensional algorithm and reduces in one dimension to a suboptimal histogram-based approach like PDPODME.

Although PRIVTM and PRIVMOM attain drastically different accuracies, it is interesting that neither improves much with a greater privacy budget, as seen in Figure 2. We believe this is because of PRIVATEQUANTILE's use of the Sparse Vector Technique, which, as explained in (Dwork et al., 2014), incurs a privacy loss only for the query values that are near or above the threshold instead of incurring a loss on every query. This more economical use of the privacy budget enables PRIVMOM to attain rates much closer to the optimal bound than methods using noisy means with noise calibrated to comparatively conservative sensitivity bounds.

²We select hyperparameters of each method non-privately here to compare the "best" version of each algorithm and are not suggesting this be done in practice (though we did find that they do not affect performance much. See Appendix B for detail).


 Figure 2: Performance of mean estimators with varying ϵ .

3.3 HYPOTHESIS TESTING

By Figure 1c, PRIVMOM does not outperform the best theoretically achievable error rate by a private mean estimator. However, skeptical of its stunning improvement over other private mean estimators, we attempt to check whether its performance exceeds statistical hypothesis testing bounds. We construct a dataset X , then construct X' by inserting a canary data point. We plot the distributions of $\text{PRIVMOM}(X)$ and $\text{PRIVMOM}(X')$ and plot the error curve of the likelihood ratio test for membership inference by varying the threshold.

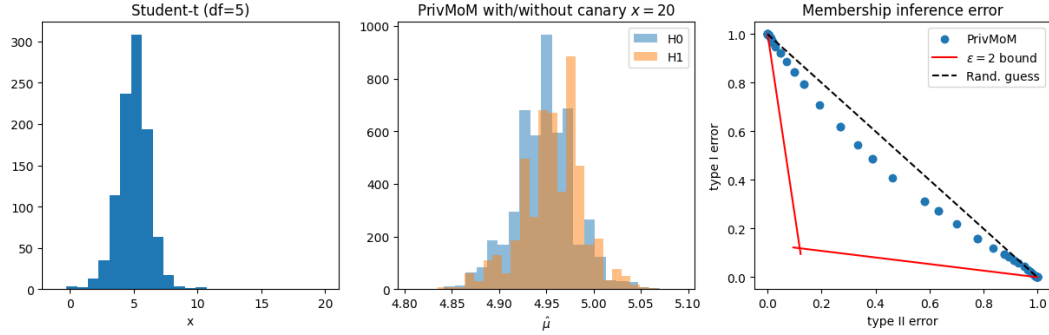


Figure 3: Distributions of the means produced by PRIVMOM in 5000 simulations (during which the dataset on the left hand side remained fixed). H_1 contains the canary value $x = 20$. The Student-t dataset contains 1000 points; PRIVMOM performs poorly with $N < 1000$, and with more points, it becomes increasingly difficult to discern the presence of any individual value. Other attempts at membership inference hypothesis testing can be found in Appendix C.

We conducted many experiments with various datasets and canaries, with the best likelihood ratio result we could obtain shown in Figure 3. This does not constitute proof that PRIVMOM does not violate theoretical bounds, but it does nicely demonstrate its robustness to singular outliers.

4 CONCLUSION

We have implemented differentially private versions of the classic median of means and trimmed mean of means for robust mean estimation. We have shown PRIVMOM's superior performance to other private mean estimation algorithms and have attempted to check that it does not exceed theoretically achievable error rates.

4.1 FUTURE WORK

A careful theoretical analysis must be conducted to ensure that our proposed PRIVMOM does not violate the heavy-tailed minimax rate put forth in (Kamath et al., 2020) because PRIVMOM vastly outperforms PDODME, which is minimax optimal. PRIVMOM does not have an α parameter, but it is needed to calculate the minimax rate from Kamath et al. (2020).

On the more practical front, our implementation of PRIVMOM is not numerically stable. See Appendix D for examples of instability. As shown by (Mironov, 2012), implementations of differentially private algorithms with floating point number systems and numerical instability can cause total privacy loss. A good place to start may be to improve the choice of candidate thresholds as opposed to exponentially incrementing β^i , or even initializing β to $1.001 \pm [\text{noise}]$. It would also be interesting to try initializing the lower bound in PRIVATEQUANTILE to a very rough private estimate of the quantile instead of 0, as PRIVMOM seems to do poorly when the quantile is very far from 0.

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A APPENDIX

Algorithm 3 is from Section 3.6 of Dwork et al. (2014). Algorithm 4 is from Durfee (2023).

Algorithm 3 ABOVETHRESHOLD

```

1: Input:  $X$ , queries  $\{f_i : \mathcal{X} \rightarrow \mathbb{R}\}$  with sensitivity  $\Delta$ , threshold  $T$ ,  $\epsilon_1$ ,  $\epsilon_2$ 
2:  $\hat{T} \leftarrow T + \text{Lap}(\Delta/\epsilon_1)$ 
3: for each query  $i$  do
4:   if  $f_i(X) + \text{Lap}(\Delta/\epsilon_2) \geq \hat{T}$ , return  $\top$  and halt
5:   else return  $\perp$ 
6: end for

```

Algorithm 4 PRIVATEQUANTILE

```

1: Input:  $X$ , quantile  $q$ ,  $\epsilon$ 
2: Run ABOVETHRESHOLD with  $X$ ,  $T = qN$ , and  $f_i(X) = |\{x_j \in X : x_j + 1 < \beta^i\}|$ 
3: Run ABOVETHRESHOLD with  $X$ ,  $T = (1 - q)N$ , and  $f_i(X) = |\{x_j \in X : x_j - 1 > -\beta^i\}|$ 
4: if the first ABOVETHRESHOLD halts with  $k > 0$  iterations, return  $\beta^k - 1$ 
5: if the second ABOVETHRESHOLD halts with  $k > 0$  iterations, return  $-\beta^k + 1$ 
6: else return 0

```

B HYPERPARAMETER TUNING

See Figure 4 for PRIVMOM hyperparameter tuning. We find that neither the way that the privacy budget is split between ϵ_1 and ϵ_2 , nor the number of blocks B , makes a significant impact on the estimator’s accuracy. Thus we choose $\epsilon_1 = \epsilon_2 = \epsilon/2$ and $B = 200$.

See Figure 5 for PDPODME hyperparameter tuning. We find that R , the range within which μ is presumed to lie, makes no difference in performance.

C MEMBERSHIP INFERENCE HYPOTHESIS TESTING

See Figures 6, 7, 8 for other attempts.

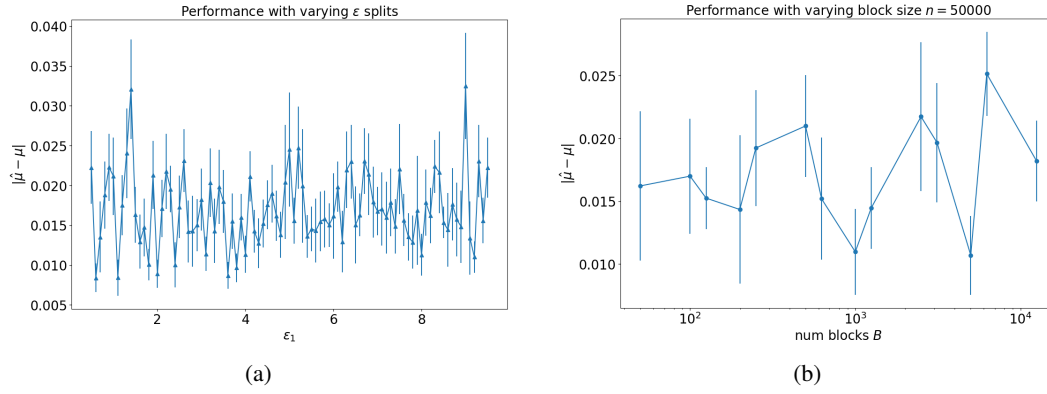
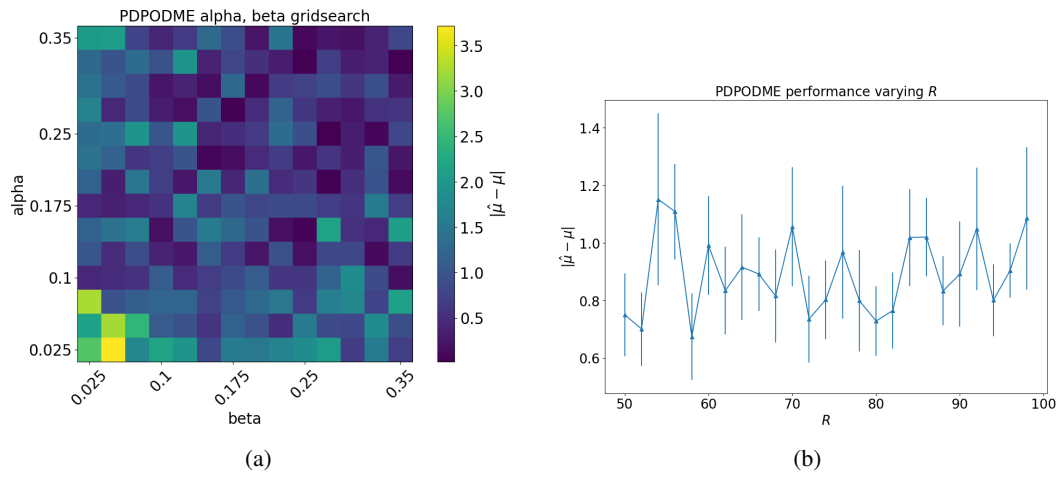


Figure 4


 Figure 5: α, β, R gridsearch for PDPODME.

D NUMERICAL INSTABILITY

In Figure 9, repeatedly inputting a dataset consisting of half 0's and half 1's into PRIVMOM does not result in a smooth distribution of means around 0.5.

In Figure 10, because β^i is incremented in a discrete manner, output $\hat{\mu}$ fall into discrete bins.

In Figure 11 and 12, β^i begins to increment by larger and larger margins to reach large $\hat{\mu}$'s.

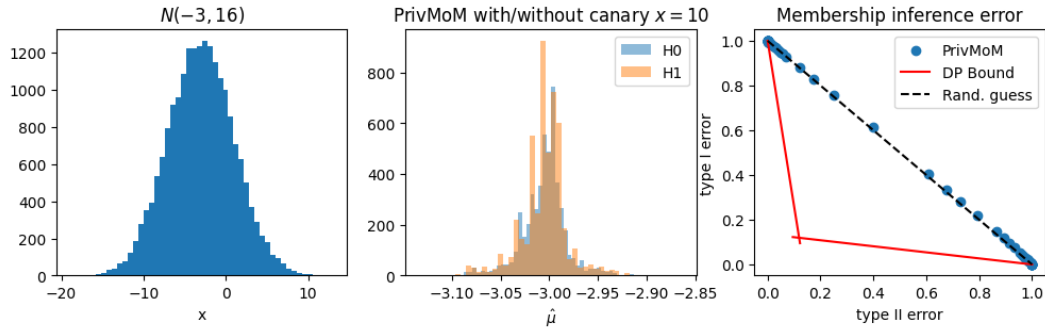


Figure 6

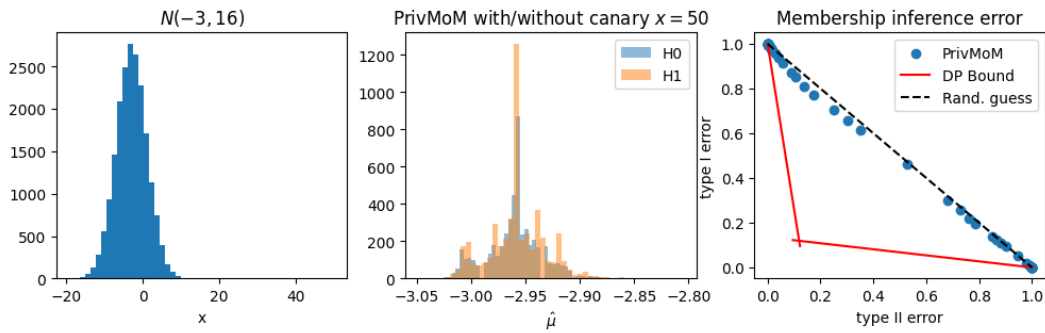


Figure 7

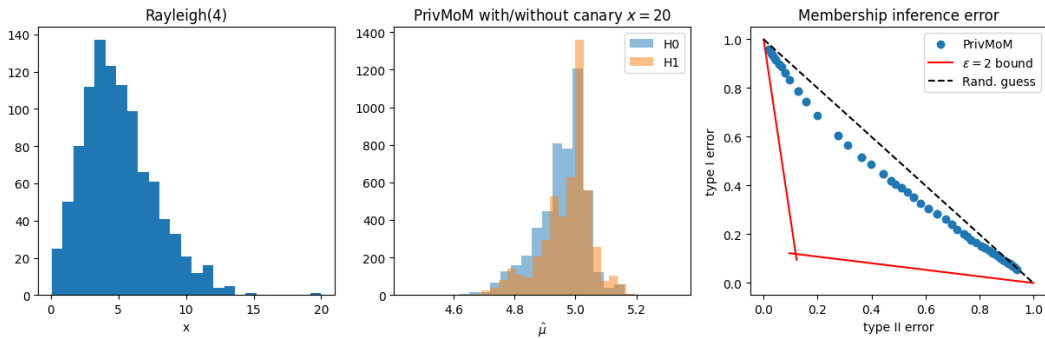


Figure 8

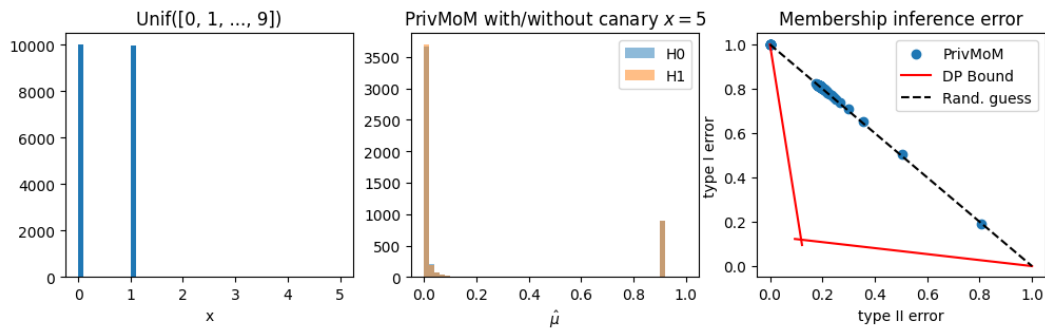


Figure 9: Certain $\hat{\mu}$'s are never output by PRIVMoM while others are at abnormally high frequency.

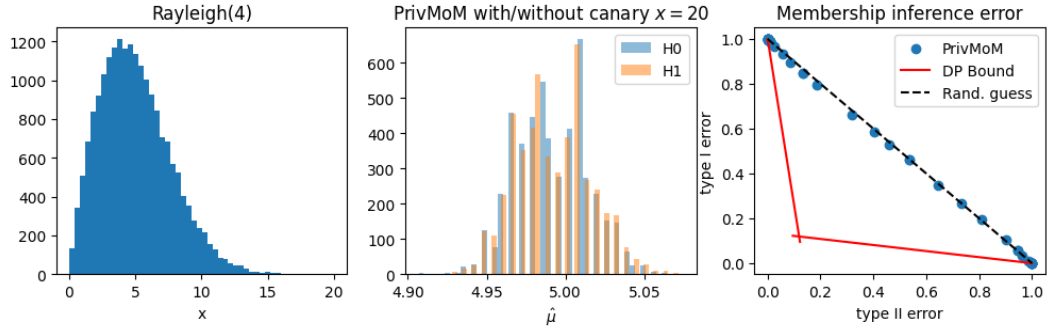


Figure 10: The output of PRIVMOM falls into discrete bins.

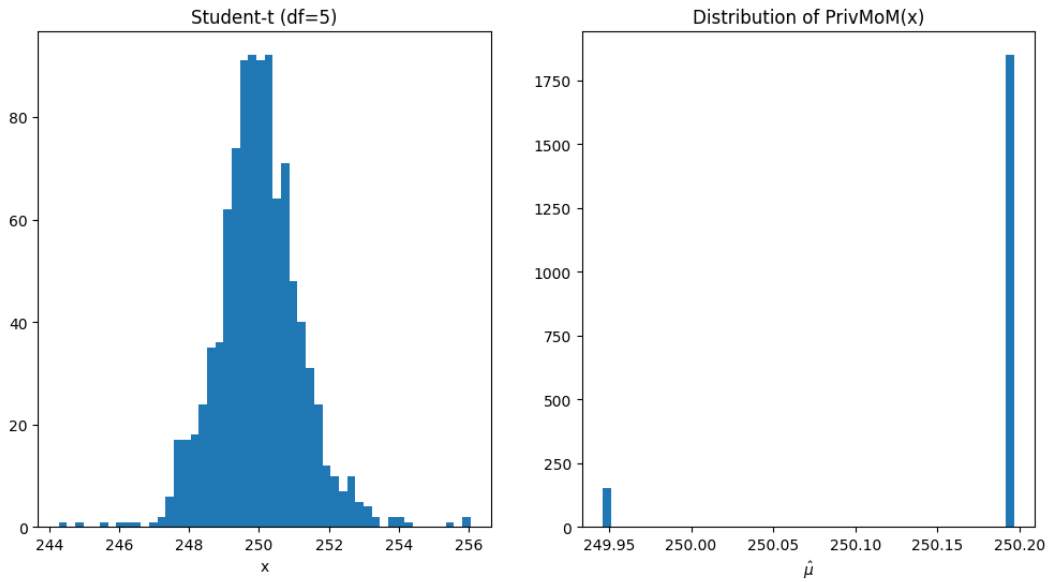


Figure 11: The output of PRIVMOM with large distribution mean

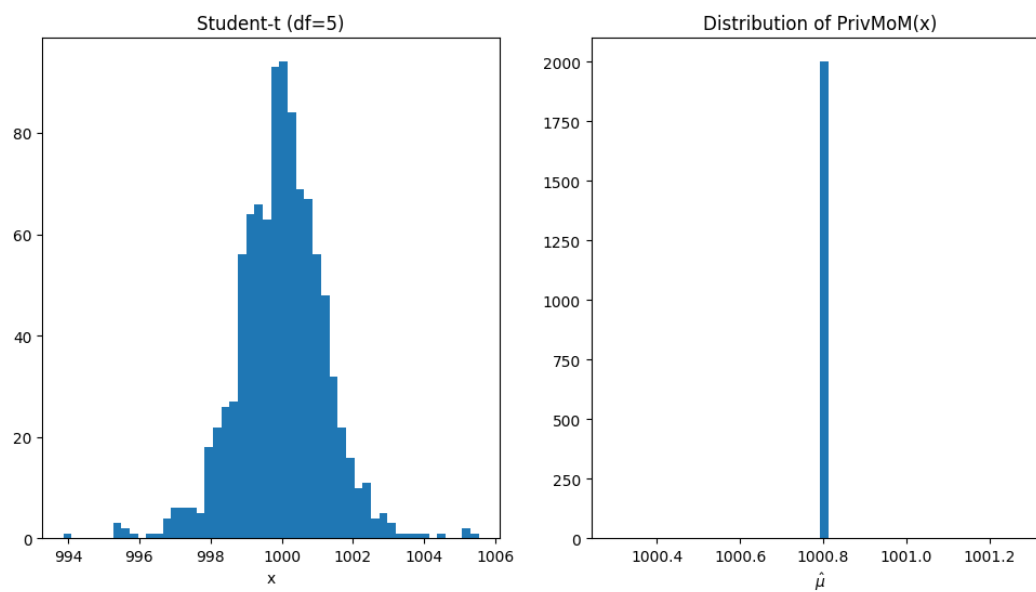


Figure 12: The output of PRIVMOM with large distribution mean